Tensagrity - Casuals Project Report

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Objective

Build an untethered, motor-actuated tensegrity robot that moves by changing its own shape, survives the impacts that come with rolling and falling, and operates via wireless communication. The project follows Yale's open-source tensegrity robot as a reference, adapted to the team's own electronic optimizations. To successfully meet the objective of this project, the robot must be able to:

1. Move by shape change: Shorten and lengthen its actuated cables with motor-driven spools to adjust tension, shifting its center of mass so it tips and rolls from one face to another.
2. Steer towards a goal: Choose which cables to actuate so that it can roll in a chosen direction and travel towards a target location.
3. Hold its shape under load: Keep every cable in tension so that struts stay separated and the structure stays stable while it stands, rolls, and carries its internal components
4. Absorb impacts: Survive drops from increasing heights and rolling tip-overs without the structural body or electronics breaking
5. Sense its own state: Measure strut orientation, cable length, and power draw so the controlling unit knows how the robot is oriented and configured and can proceed towards a goal.
6. Operate untethered: Run on onboard batteries and operator over Wi-Fi connections from an external controller, with no cables to a base station.

Available designs

Yale - 3 strut design

The first available design that is being considered is deemed the Yale design. This design would involve constructing a product similar to the one produced by the Yale group. This design would capitalize on the easy open-source replicability and insight to the design process. This would prove to be a great benefit in preparing a functional design in the allotted 10 week period. The design works by using only 3 struts and having a winch attached to a motor at the end of each strut to pull the robot. An image of the constructed Yale design can be seen in Figure 1 below.

Some negatives for selecting this design would be that the structure would be a little weaker and that there would be complexity with the locomotive configuration. The other more personal group negative would be that this design would lack some originality and be reflective of the performance of the Yale design.

Simplified pros and cons of the design are listed below:

Pros:

- Short local wiring
- Reusable modules/components
- Easy open sourced replicability

Cons:
- Structural weakness with increasing drop height
- Complex programming between free-fall and locomotive configurations

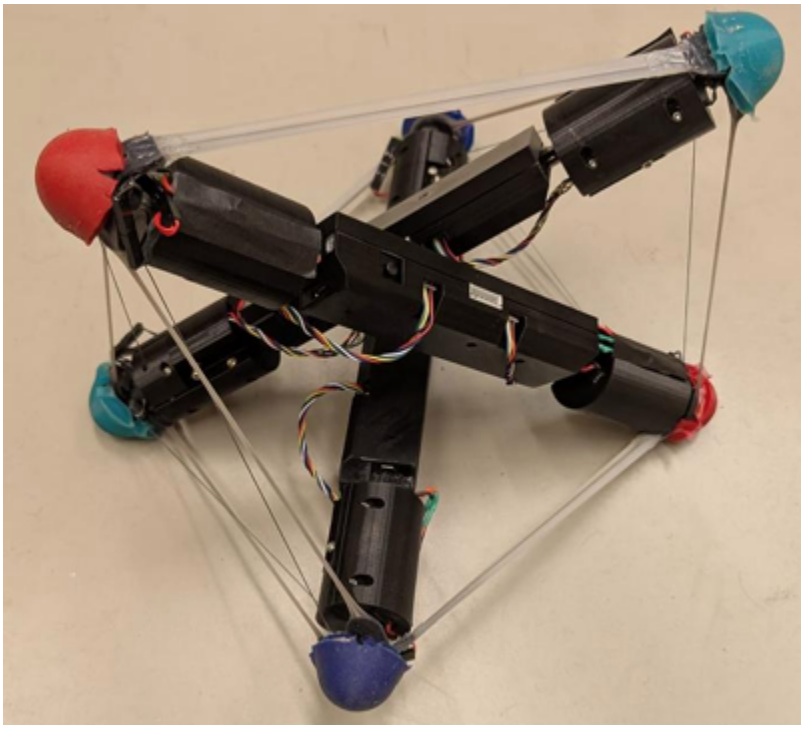


Figure 1: Yale 3 Strut Design

Central controller design

An alternate design that was evaluated involved moving the controls and electronics to the center of the robot as seen in Figure 2. This would allow for larger electronic boards that would not be limited by the diameter of the struts. This design also allows for the electronics to be a protected central mass, protecting the electronics (ideally) from long drops to the bottom. Moreover, the design allows for built-in structural integrity as opposed to the previous three-strut design.

Some negatives of the design revolve around the uncertainty of how the design process will go. This design would be a more intrinsically interesting design as it would be something different from the Yale design, which is very well documented. However, in this uncertainty there lies a concern that the design would be hard to construct and program in 10 weeks. The natural design of the design would involve more complex mechanical design and increased design variables.
Pros:
- Protected central mass

- Design allows built-in structural integrity - does not need “free-falling” configuration
- Simpler control system due to central controller

Cons:

- More complex mechanical design
- Complex electronics and power design needs to be done from scratch
- Increased design variables - suspension, inertia, moving wire routes

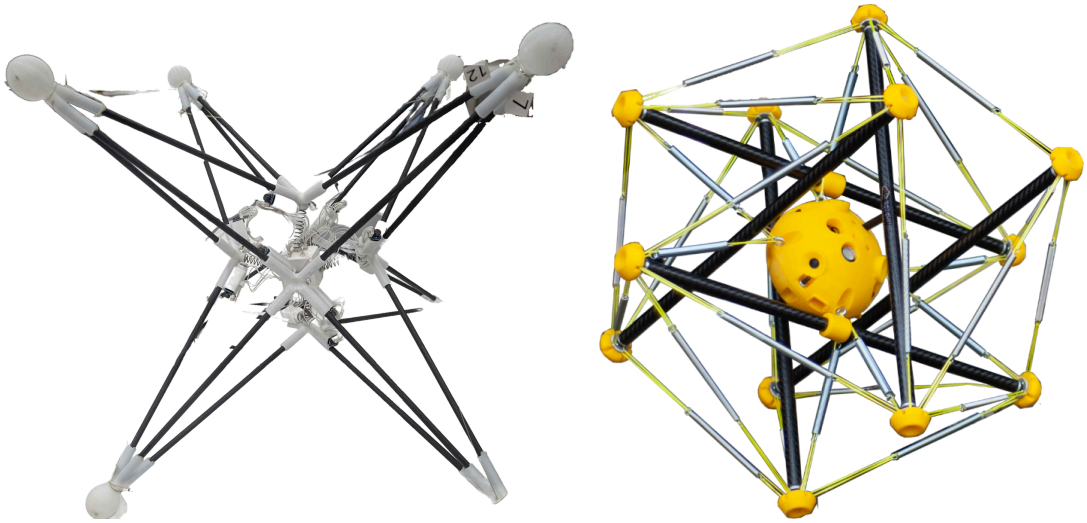


Figure 2: Central Mass Design Images

Concept Selection Design

Ultimately, due to time constraints and practicality of design, Yale’s mechanical design will be replicated for this challenge. The availability of the resources and knowledge from the Yale design will allow for a more efficient design process as it builds on a known foundation. Thus, the design process will move not to constructing one from scratch but to improving and replicating an existing design. As this would be the first design the team has done with tensegrity, building on others’ experience will allow for a more coordinated and focused design approach.

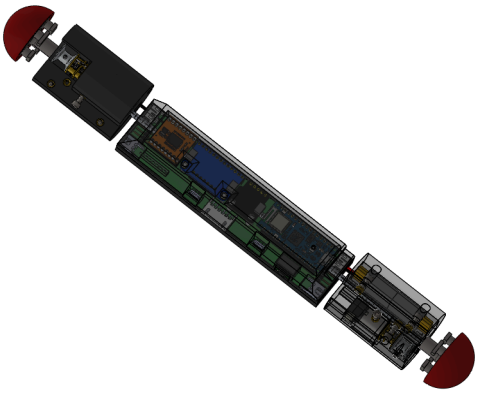


Figure 3: Image of one of the Yale Struts taken in Soldworks

Component selection

After selecting the design, the team went through and generated a list of potential components that would need to be acquired for the design. Important to note here that many components were influenced by the components the Yale group have selected for their three strut design. Below is a focused view of some of the more important components of the design. Most of these will focus on discussing the electronic components, as they will be the most important components of the design and what the mechanical components will be built around.

- Waveshare General Driver Board:
    - Seen in Figure 4
    - 65 x 65 mm physical dimensions
    - It has support for the motor controller and IMU built in; however, there are a lot of other unwanted peripherals included in the package (lidar and camera ports)
    - **Rejected due to spatial constraints**. If we were to choose this as the controller for the strut, a new mechanical design needs to be done to accommodate this board power and increased mass requirements



Figure 4: Image of the Waveshare General Driver Board.

PCB Design

The Yale main controller was modified to remove unwanted hardware components, including the force-feedback mechanism and the electromagnet drivers. Physically, the same PCB form factor and placement were maintained for backwards compatibility. Figures 5 and 6 show the schematic and the PCB layout generated by Fusion360.

Gradle files were generated using Fusion360 and sent off to fabrication on 16/10/2026

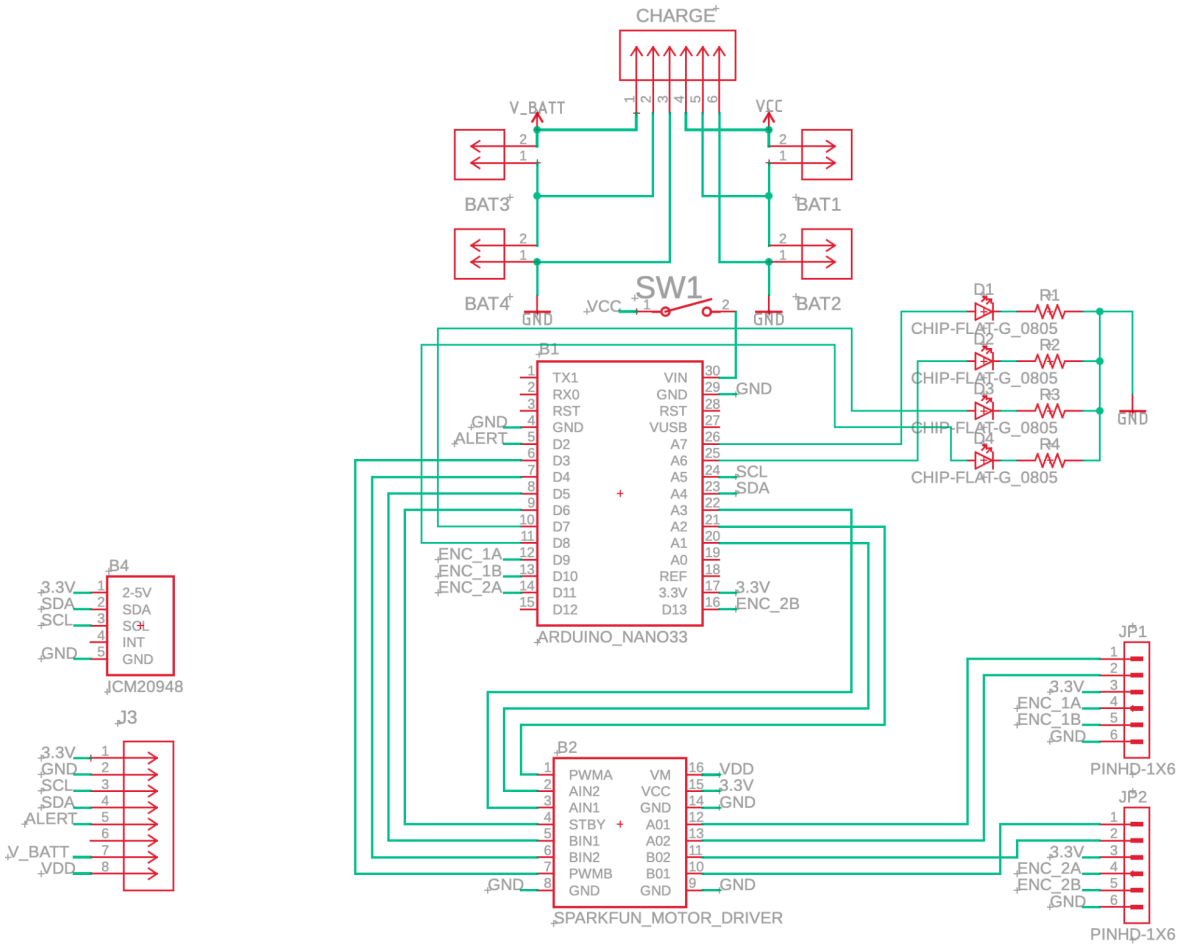


Figure 5: Modified PCB design after removing unwanted components

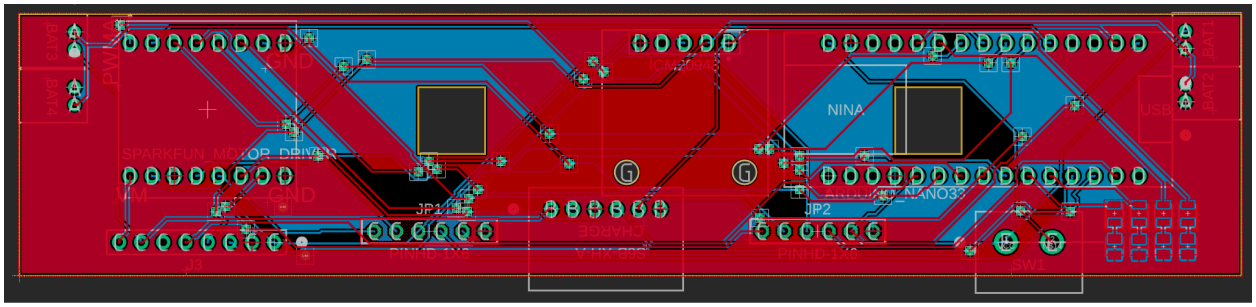


Figure 6: PCB design maintaining the original dimensions and all physical constraints

High-level wiring diagram

Power is split into two independent 2S LiPo packs. The logic pack feeds the Arduino through the power switch, and the Arduino's regulator supplies 3.3 V to the IMU, power monitor and motor-driver logic. The motor pack feeds the motor driver through the INA260, so the controller can measure motor power draw. The Arduino sends PWM and direction signals to the driver and reads both motor encoders directly.

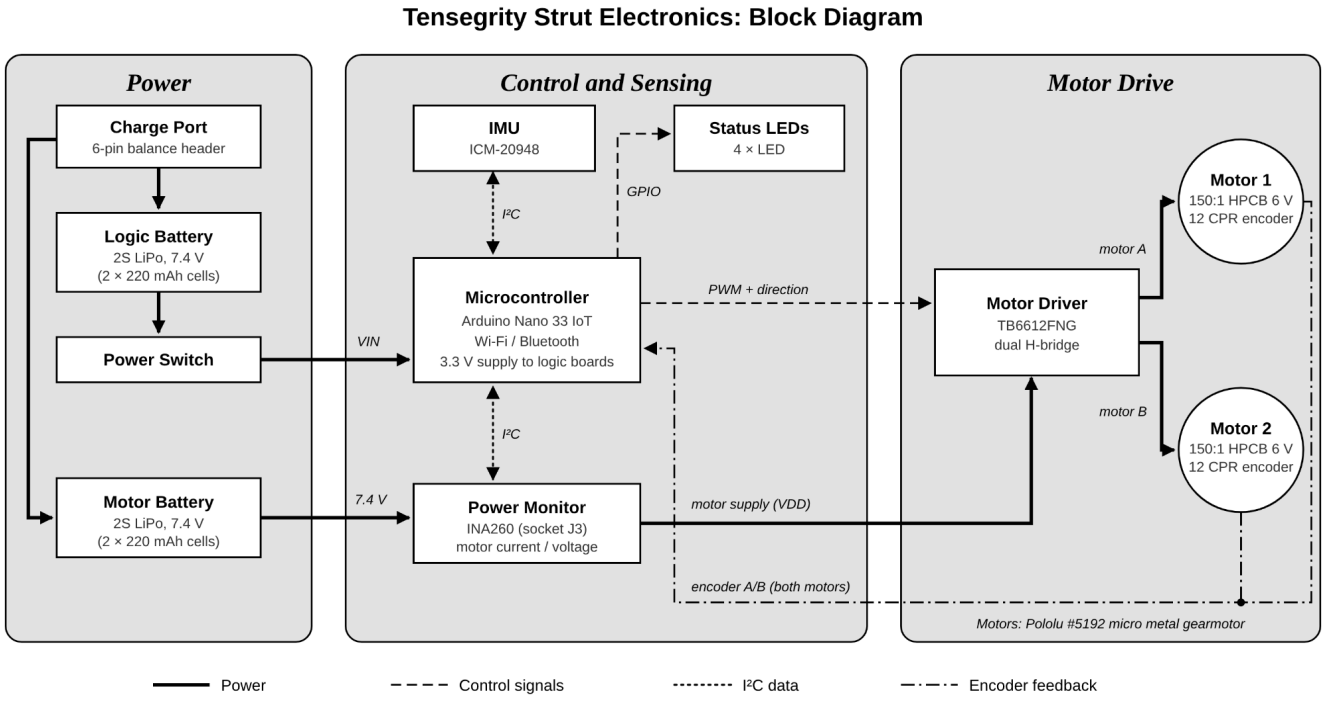


Figure 7: Block diagram of one strut's electronics. All three struts carry an identical copy.

Control system design considerations

The Yale design is designed to function as three independent struts with their own inputs and outputs. Our control system design would follow the same principles. We plan to use wi-fi and MQTT messages as a way to control the individual struts. This would be considered stage 1 implementation. Here, external human inputs will be used as feedback to maintain the robot's shape. We define a minimum success criterion for stage 1 as being able to move under a recorded sequence of moves

After basic strut controller design and testing, we may choose to use the Robot Operating System as a way to implement detailed control algorithms. For this purpose, we would like to buy a Raspberry Pi 4 with 4GB of RAM. This would not be immediately needed but would be something that we would implement as part of stage 2.

Time line

Description	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Proposal and BOM												
PCB design												
Electronics testing												
Mechanical building												
Make 2 copies												
Stage 1 control system												
Drop testing												
Movement												
Stage 2 control system (ROS)												

Note: Green boxes are tasks that are already being completed; blue indicates tasks that are pending in the future

BOM

After choosing to pursue the construction of the three-strut design by Yale, the Bill of Materials was constructed to determine the cost of the project. Since we are using the Yale design, the bill of materials for the robot consisted of the same or similar components. The mechanical components consisted mainly of screws or bolts with the housing for the design being designed with CAD and produced using FDM printing available.

Component	Quantity	Unit Price	Total	Supplier	order stages	872.28
PCB	1	$49.00	$49.00	Other	Stage 0	already ordered
Pin headers (40 pin) 10 pack	1	$5.99	$5.99	Amazon	Stage 1	
LiPo Battery 220 mAh with jumper (6 pack)	2	$26.99	$53.98	Amazon	Stage 1	
JST 6pin connector - 5 pack	1	$8.99	$8.99	Amazon	Stage 1	
Arduino Nano 33 IoT	4	$23.90	$95.60	Arduino	Stage 1	One extra controller for backup
INA260	3	$9.95	$29.85	Digikey	Stage 1	needed - possible feedback
JST S2B (battery connector 4 per PCB x 3)	15	$0.09	$1.35	Digikey	Stage 1	
S6B connector - charger	3	$0.32	$0.96	Digikey	Stage 1	
Mechanical components (sum mirror of Yale)	1	$130.00	$130.00	Other	Stage 1	
Micro Metal Gearmotor HPCB 6V w/ Encoder	6	$32.45	$194.70	Other	Stage 1	
Battery Charger for 2S 2 cells	1	$31.99	$31.99	Other	Stage 1	
Dual TB6612FNG Motor Driver	3	$14.50	$43.50	Sparkfun	Stage 1	
PIM448 - IMU	3	$18.79	$56.37	Digikey	Stage	

					1/2	
Raspi - 4	1	$120.00	$120.00	Adafruit	Stage 2	optional - but prefered
Miscellaneous (strings, and safety margin for any changes)	1	$50	$50.00	Other	Stage 1	

Detailed spreadsheet with links can be found here
All Digikey orders are in a list for easy order: https://www.digikey.com/en/mylists/list/HJK59BJTE2

Documentation

All documentation will be maintained in a GitHub repo for easy contribution and maintenance

Appendix

1. Images of the costs of the components as seen at the carts of the respective websites where the components are being purchased.



Figure 8: Cost of mechanical components bought from McMaster